Multimodal Optimization of Piezoelectric Patch Placement and Shunt Circuit Design for Enhanced Structural Vibration Control

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Abstract

Composite materials have become increasingly prevalent in industrial applications, replacing traditional materials due to their lower structural mass and superior mechanical properties. However, these materials are often more susceptible to structural vibrations, which can degrade performance and reduce operational lifespan. To address this issue, passive control techniques such as piezoelectric shunt damping have gained attention for their simplicity and effectiveness. This method converts vibrational mechanical energy into electrical energy through piezoelectric patches, which is then dissipated via electrical circuits composed of resistors and inductors. Nevertheless, key challenges remain, including the optimal positioning of the patches and tuning of the circuit parameters. This thesis presents a numerical model developed in ANSYS to optimize the application of piezoelectric patches on composite plates. Two optimization strategies were investigated: a unimodal approach focusing on the dominant vibration mode, and a multimodal approach considering all excited modes within the frequency range of interest. Optimization was carried out using the Global and Local Optimization using Direct Search (GLOS) and Direct Multisearch (DMS) algorithms. The results enabled a comparative assessment of the effectiveness of each control strategy and provided insights into the conditions under which each is most suitable. Therefore, the study offers practical guidelines for the efficient implementation of shunt control in composite structures subjected to complex vibrational excitations.

Keywords: Piezoelectric materials, Passive shunt control, Electromechanical coupling coefficient, ANSYS, Multimodal optimization

1. Introduction

The use of composite materials [5] as an alternative to metals has become increasingly common in industrial applications, driven by the demand for higher structural performance, robustness, and competitiveness of final products. Composites combine the best properties of their constituents within a single structure, making them especially attractive in sectors with stringent technological requirements. The aerospace industry, for instance, has recognized the potential of composites since the 1990s [3]. More recently, the automotive and robotics sectors have also expanded their adoption, leveraging the light weight, precision, and efficiency of composites to achieve significant engineering advancements. This adoption highlights the versatility of composites and reinforces their strategic role in enabling next-generation engineering solutions.

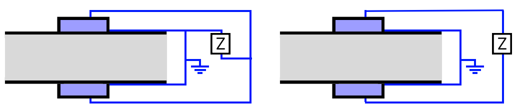
Despite these advantages, composites exhibit structural limitations, particularly in environments subject to high vibration levels, which can compromise both stability and service life. Repeated dynamic loading can accelerate fatigue mechanisms, induce premature failures, and even lead to resonance-related instabilities that jeopardize safety. To mitigate these problems, smart-material-based solutions have emerged. These combine composites with piezoelectric elements, which can act simultaneously as sensors and actuators. Among the possible strategies, passive shunt damping stands out: in this approach, resistors and inductors are connected to piezoelectric patches, enabling the conversion of mechanical vibration energy into electrical energy. This energy is then dissipated by the circuit, resulting in reduced vibration amplitudes, suppression of unwanted resonances, and improved structural reliability. Furthermore, the passive nature of shunt damping eliminates the need for continuous external power supply, making it a cost-effective, lightweight, and low-maintenance solution when compared to active vibration control systems.

However, the application of shunt damping involves significant challenges from electrical (such as equipotential interfaces, shunt-type connections, and resistor and inductor values), geometrical (placement and geometry of the patches), and mechanical perspectives (such as materials, boundary conditions, and electromechanical coupling).

On the mechanical side, the focus is on calculating the electromechanical coupling coefficient, which determines the effectiveness of energy conversion between mechanical and electrical forms in piezoelectric materials. This concept was first introduced as the ElectroMechanical Coupling Coefficient (EMCC) in [4], and since then, several formulas have been developed to calculate this coupling effect, highlighting the quasi-static EMCC [6] and the effective EMCC [8, 2].

On the electrical side, the first challenge lies in defining the configuration of the smart circuit, which depends on the following aspects:

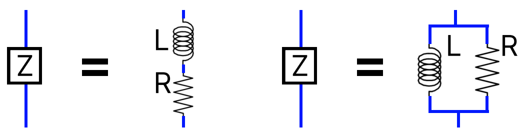
- **Conducting plate structure**



(a) Same Polarity Parallel Connection (SP-PW). (b) Opposite Polarity Series Connection (OP-SW).

Figure 1: Conducting plate structure with a pair of patches wired for a shunt circuit.

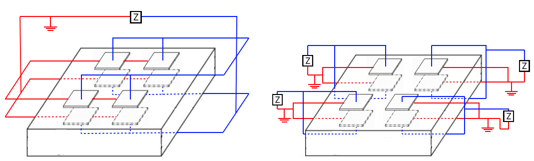
- **Shunt circuit topology**



(a) Series shunt circuit. (b) Parallel shunt circuit.

Figure 2: Shunt circuit configurations connected to a pair of patches.

- **Number of electrical circuits**



(a) Single shunt configuration. (b) Multi shunt configuration.

Figure 3: Configurations varying the number of shunt circuits connected to the structure under SP-PW conduction.

Once these configurations are established, the remaining challenge lies in identifying the optimal resistor and inductor values to achieve effective vibration mitigation.

Therefore, the scope of this thesis is to develop a numerical model capable of identifying the optimal geometric and electrical characteristics to support the operational demands of the composite material's application under passive control. This thesis aims to create a self-responsive multilayer material, requiring no active actuators, that not only delivers the structural advantages previously outlined but also maintains stability and durability in dynamic application environments.

2. Piezoelectric-Plate Model

The analysis of electromechanical coupling in smart structures involves several complexities, requiring the adoption of appropriate theoretical and numerical models to accurately represent their behavior. In this thesis, three complementary models were employed: a three-dimensional theoretical model, which describes the physical foundations of piezoelectric coupling; a numerical model in ANSYS developed to simulate different spatial configurations of the system; and an optimization model, aimed at determining the optimal solutions for the various control approaches investigated throughout the work.

2.1. Theoretical Model

In the literature, theoretical representations of piezoelectric structures are generally divided into two main approaches: two-dimensional and three-dimensional. The two-dimensional approach introduces simplifications that describe mechanical and electrical behavior through averaged stress–strain relationships, reducing computational cost. The three-dimensional approach, in contrast, is based on solid mechanics equations, allowing a more accurate capture of local deformation and electromechanical coupling effects.

In this work, the three-dimensional approach was adopted since the numerical model was implemented using ANSYS software, which employs the Finite Element Method (FEM). This methodology enables precise computation of displacement, strain, and electric potential fields while accounting for the anisotropic properties of materials, boundary conditions, and applied loads. Consequently, it provides a realistic representation of the dynamic behavior of the plate and the bonded piezoelectric patches.

In addition to the dynamic characterization, a detailed analysis of the Electromechanical Coupling Coefficient (EMCC) was performed — a key quantity for evaluating the efficiency of energy conversion between the mechanical and electrical domains. As discussed in Section 1, different formulations exist for computing this coefficient. However, for the scope of this research, the Effective EMCC model was adopted due to its accuracy and ease of implementation in ANSYS, since the software directly provides the natural frequencies of the system under different electrical conditions. The Effective EMCC is defined as follows:

$$k^2_{e(m)} = \frac{\omega^2_{OC(m)} - \omega^2_{SC(m)}}{\omega^2_{SC(m)}} \tag{1}$$

where $\omega_{OC(m)}$ represents the natural frequency of mode m under open-circuit conditions ($Z = \infty$), characterized by the absence of electric charge ($\mathbf{Q} = 0$). Conversely, $\omega_{SC(m)}$ denotes the natural frequency under short-circuit conditions ($Z = 0$), when $\mathbf{Q} \neq 0$. The value of $k^2_{e(m)}$ is typically expressed as a percentage and provides a direct measure of the coupling strength between the mechanical and electrical energies of the system.

Another fundamental theoretical model adopted in this research concerns the calibration equations of the electrical elements within the control circuit. This model is essential since one of the main results presented in this thesis relates to the implementation of a single-shunt circuit tuned through optimized electrical parameters (resistance and inductance), aiming to achieve an effective reduction of the mechanical vibrations of the structure.

Among the various RL circuit calibration models studied, the one based on the tuning procedure described by [7] stands out. This approach is widely used in the literature due to its strong correlation between the theoretical and experimental behavior of piezoelectric systems. The method defines analytical expressions that directly relate the electromechanical coupling coefficient ($k_{e(m)}$), the short-circuit natural frequency ($\omega_{SC(m)}$), and the modal charge ($Q_{SC(m)}$) to the optimal values of resistance (R) and inductance (L), ensuring efficient tuning between the electrical circuit and the dominant vibration mode of the structure.

The resulting equations for the parallel circuits are given as follows:

$$R = \frac{k^2_{e(m)}\ \omega_{SC(m)}}{Q^2_{SC(m)}} \sqrt{\frac{1}{2k^2_{e(m)}}} \tag{2a}$$

$$L = \frac{k^2_{e(m)}}{Q^2_{SC(m)}} \tag{2b}$$

where $Q_{SC(m)}$ and $\omega_{SC(m)}$ represent, respectively, the modal charge and the short-circuit natural frequency for mode m.

2.2. Numerical Model

The use of finite element models to represent smart structures has become increasingly widespread, as discussed in [1]. In this context, a comprehensive numerical model was developed and implemented in the ANSYS software to simulate the mechanical, electrical, and geometric conditions considered within the scope of this thesis.

To achieve the objectives of this research, the model was designed with high flexibility, allowing the simulation of various structural configurations, material properties, and electrical circuit parameters. This adaptability enables the assessment of different control strategies and ensures that the model can be adjusted to a broad range of operating scenarios and boundary conditions.

To clarify the logic and methodology implemented in the ANSYS environment, the following sections — together with the flowchart in Figure 4 — present a step-by-step description of the process. The overall workflow is organized into three main stages:

2.2.1 Model Generation

The Model Generation stage involves the finite element pre-processing necessary to construct the structure and define boundary conditions. The numerical model was developed in ANSYS to represent various mechanical, geometric, and electrical configurations of the smart structure. Input parameters include the geometry (plate and patch dimensions, number of patches, and their positions), electrical setup (shunt circuit type, resistor and inductor values for each patch), finite element model properties (material characteristics and mesh discretization), applied loads (magnitude and location of forces), and analysis specifications for both modal and harmonic evaluations.

Once the parameters are defined, a three-dimensional finite element model is generated, including:

1. **Geometric model:** Automated creation of points, lines, areas, and volumes based on patch dimensions and positions.
2. **FEM mesh:** Structured mesh was adopted to enhance the numerical solution's speed and efficiency.
3. **Contact surfaces:** Representing patch adhesion to the plate, assuming no separation via node-to-surface contact.
4. **Equipotential surfaces:** Connected according to the SP–PW configuration, defining electrical impedance between the inner and outer surfaces of each patch pair.

Boundary conditions were applied in both electrical and mechanical domains: grounding the inner patch surfaces and imposing mechanical constraints

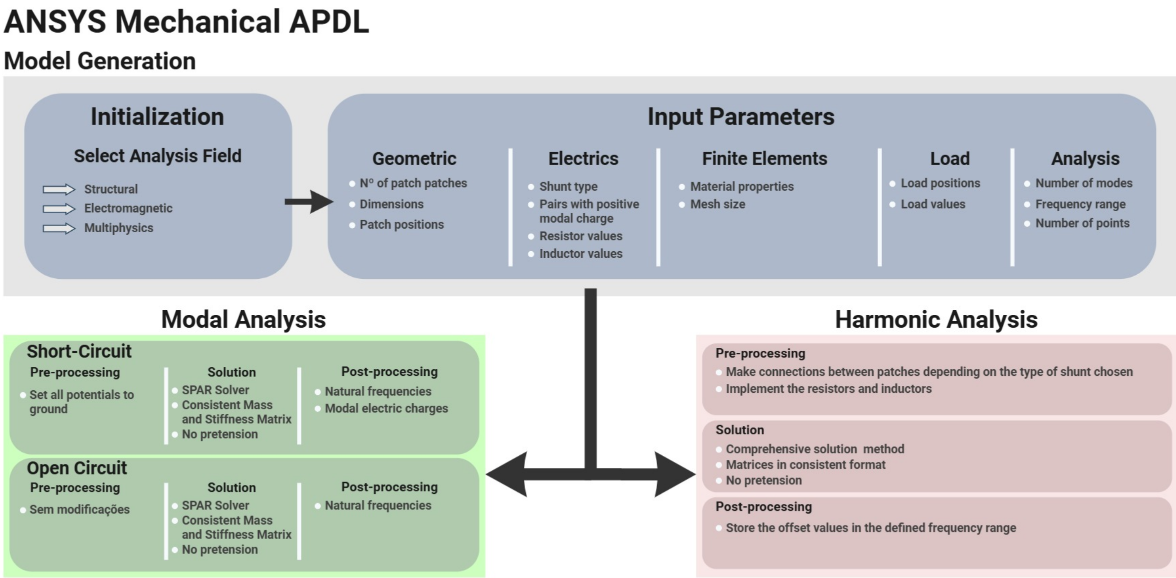


Figure 4: Flowchart of the ANSYS program implemented in this thesis.

to stabilize the plate, including vertical displacement restrictions along the edges and full constraint at two corners to fully define the problem.

2.2.2 Modal Analysis

After the finite element model is defined, modal analyses are conducted under both open- and short-circuit conditions to determine the system's natural frequencies and the effective EMCC (Eq. 1).

In the open-circuit condition, the patch terminals remain electrically disconnected, and the analysis is performed using the Sparse Matrix Solver to ensure accurate mode shapes from consistent mass and stiffness matrices. For the short-circuit case, the external and internal patch electrodes are grounded, imposing a zero-impedance boundary. The same solution procedure is applied, and the resulting natural frequencies and modal charges are extracted. These quantities—especially the modal charges—are crucial for correctly coupling the electrical and mechanical domains in subsequent single- and multi-shunt designs. Thus, the modal analysis provides the fundamental parameters for control design: natural frequencies, effective EMCC, and modal charges.

2.2.3 Harmonic Analysis

The harmonic analysis assesses vibration attenuation when resistive–inductive (RL) circuits are connected to the piezoelectric patches. In the single-shunt configuration, all patches share the same R and L values, whereas in the multi-shunt setup, each patch pair is assigned an independent circuit. External harmonic loads are applied with defined magnitude and location, and a parallel-shunt topology is adopted for all cases. The structural response is computed across the frequency range using a consistent matrix formulation to capture dynamic coupling effects. From the solution, the displacement amplitude at the excitation point is obtained, enabling evaluation of the passive control performance. Consequently, harmonic analysis provides frequency-dependent displacement data that quantify the effectiveness of each control configuration in suppressing structural vibrations.

2.3. Optimization Model

Once the numerical model is capable of handling any configuration proposed in this thesis, the next step is to apply the optimization model, which will be used to find the best solutions for each of the proposed problems. As previously discussed, the ANSYS software was employed to simulate the smart structure and provide the numerical outputs required for analysis. Given that ANSYS operates as a black-box solver and the problem involves a large number of variables, search algorithms were selected as the optimization strategy. These methods, based on point-evaluation techniques, are particularly suitable for derivative-free problems and enable faster convergence when exploring large solution spaces.

Therefore, the optimization process in this thesis is divided into two main approaches: the Optimization of the Patch Pair Positions and the Optimization of the Electrical Components. Each optimization problem is described in detail in the following subsections.

Patch Pair Positions Optimization

For this optimization problem, the Direct Multi-Search (DMS) algorithm was selected. This multi-objective method was implemented to address the simultaneous optimization of the number and spatial positioning of the piezoelectric patch pairs on the plate. The objective functions are based on maximizing the Electromechanical Coupling Coefficient (EMCC) and the number of patch pairs, forming the Pareto front of optimal trade-off solutions.

In this approach, the domain of possible patch locations is discretized through a percentage-based normalization, allowing a general and scalable optimization framework. To ensure physically valid solutions, a MATLAB routine checks and corrects any overlap between patches before each simulation, guaranteeing that the configurations evaluated in ANSYS remain numerically stable and feasible (Figure 5).

Electrical Components Optimization

After obtaining the optimal distribution of patch pairs, the next step focuses on the optimization of the electrical circuit parameters. For this task, the Global and Local Optimization using Direct Search (GLODS) algorithm was used. This single-objective method aims to minimize the vibration peak observed in the harmonic response by adjusting the resistance and inductance values of the shunt circuits.

Two electrical configurations were considered: the single shunt, where all patches are connected to a common circuit, and the multi shunt, where each patch pair has its own circuit. The optimization is performed in the harmonic domain, using ANSYS simulations to evaluate each configuration and determine the optimal set of electrical parameters that lead to the most effective vibration reduction.

3. Implementation

With all models properly defined, it becomes essential to describe how each of them is practically implemented. This section outlines the required input parameters for both the numerical and optimization models to operate efficiently and generate consistent results aligned with the research objectives.

3.1. ANSYS Implementation

For the numerical model, the implementation was carried out using the ANSYS software, which served as the main platform for simulating the smart structure. In this stage, all fixed parameters of the problem are defined, including material properties, finite element types, geometric dimensions, mesh size, and loading conditions. These parameters remain constant throughout all analyses, ensuring consistency across different configurations.

The variable parameters, on the other hand, are defined according to the objectives of each simulation and can be grouped into two main categories: (i) optimization variables — such as the number and position of patch pairs, as well as the resistor and inductor values — and (ii) user-defined parameters, including the circuit configuration, load position, and frequency range of the analysis.

The communication between the optimization model and ANSYS is achieved through text files containing all input data for each configuration. This integration allows the model to be executed automatically and adapted to different scenarios, ensuring both efficiency and reproducibility throughout the simulation process.

Model Validation

To ensure the accuracy and reliability of the numerical model, a validation process was conducted by comparing its results with reference studies from the literature. Both modal and harmonic analyses were performed to verify the consistency of the developed model.

In the modal verification, natural frequencies and electromechanical coupling coefficients ($EMCC$) were compared under open- and short-circuit conditions. The obtained results showed negligible differences when compared with reference data, confirming the accuracy of the implemented approach.

Table 1: Comparison of modal results between the thesis and reference models.

Mode	k_e^2 [%]	$k_{e,r}^2$ [%]	**Error** [%]
1	0.41	0.41	0.00
2	1.02	1.01	0.01
3	0.94	0.93	0.01
4	1.93	1.90	0.02
5	0.63	0.62	0.02
6	0.99	0.98	0.01

As summarized in Table 1, the frequency and EMCC errors remain below 0.02%, demonstrating high numerical accuracy and confirming the reliability of the developed model.

For the harmonic verification, the frequency response of the first vibrational mode was compared with published results. The two curves exhibited almost identical behavior, with a residual sum of squares (RSS) of 6.16×10^{-4}, confirming that the proposed model can accurately reproduce the dynamic response of the system under harmonic excitation.

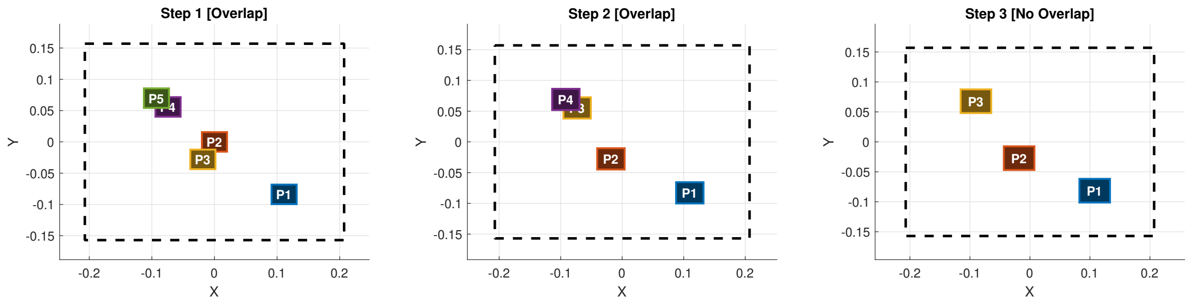


Figure 5: Example of the overlap prevention algorithm, starting with an initial percentage distribution of 5 patch pairs given by the vector **zz** = [0.25 0.75 0.30 0.70 0.50 0.50 0.45 0.40 0.80 0.20].

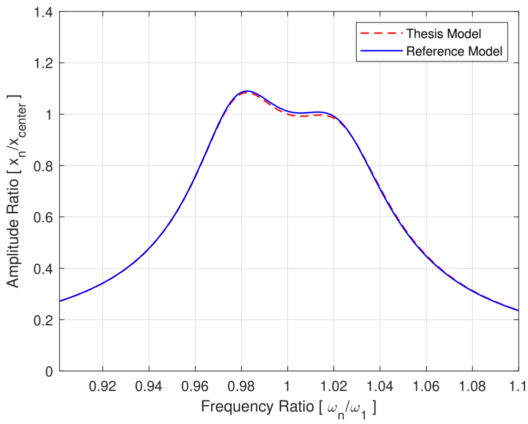


Figure 6: Harmonic verification — frequency response comparison of the thesis model with reference results.

3.2. DMS Implementation

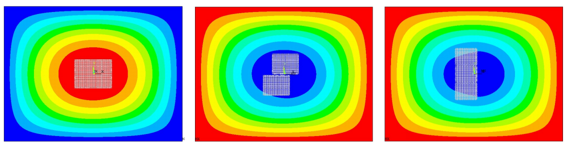
The optimization stage was implemented using the Direct MultiSearch (DMS) algorithm, a derivative-free, point-based method suitable for complex multi-objective problems such as the optimization of smart structures. This algorithm was integrated with the ANSYS numerical model to determine two main sets of optimal variables: the spatial distribution of piezoelectric patch pairs and the electrical parameters of the shunt circuit (resistance and inductance).

In this approach, each candidate solution generated by the DMS algorithm was automatically evaluated in ANSYS, with the program reading input text files containing geometric and electrical parameters. The DMS then adjusted these parameters iteratively, balancing global exploration and local refinement to efficiently approach the Pareto front. Its point-based nature reduced the computational cost compared to traditional population-based methods, making it suitable for problems involving multiple coupled physics.

For the optimization of patch positions, the algorithm handled variable numbers of patch pairs by defining inactive coordinates as $(-1, -1)$, ensuring flexibility while maintaining a fixed search dimension. This strategy enabled the model to automatically adapt to different configurations without additional preprocessing.

Initial Analysis of the Model

To validate the optimization implementation, the model was first tested using the first vibration mode, where the maximum amplitude occurs at the plate's center. The algorithm correctly positioned the patch pairs in this central region for all tested configurations, confirming its accuracy and consistency. These results demonstrated that the DMS-based approach successfully coupled with the finite element model, providing a robust framework for identifying optimal structural and electrical configurations with high precision.



(a) 1 patch pair. (b) 2 patch pairs. (c) 3 patch pairs.

Figure 7: Model validation of the optimization for the first vibration mode using up to 3 patch pairs applied to the plate.

3.3. GLODS Implementation

The optimization of the electrical parameters was performed using the Globalized Orthogonal Direct Search (GLODS) algorithm, a derivative-free global search method adapted for single-objective problems. Compared to the DMS implementation, GLODS operates more straightforwardly since it does not require constraint handling or infeasibility filtering. In this context, the optimization variables—resistance (R) and inductance (L)—are treated as continuous parameters bounded only by their admissible physical ranges.

The GLODS algorithm combines global sampling and local refinement through alternating search and poll stages. Initially, several local searches are launched from distributed points within the search space (using random or quasi-random sampling). Promising regions are then refined iteratively, while overlapping searches are merged to reduce redundancy. This hybrid structure allows the method to efficiently converge to the global optimum with a limited number of function evaluations, making it suitable for computationally intensive finite element evaluations.

In the implementation, the geometric configuration of the patches was fixed based on prior optimization results, allowing GLODS to focus exclusively on tuning the electrical shunt circuit parameters. The optimization process followed a two-step discretization scheme: first, a coarse search to identify promising intervals for R and L, followed by a refined search around the best solutions. The program automatically reused optimal points from the first stage as initial candidates for the second, improving convergence efficiency.

Initial Analysis of the Model

To validate the implementation, a test was performed on a single patch tuned to the first vibration mode of the plate. The search space was defined as $R \in [42, 47]$ kΩ and $L \in [3, 6]$ H, with discretizations of 100 Ω and 0.1 H, respectively. The algorithm converged to $R = 46$ kΩ and $L = 3.65$ H, resulting in clear attenuation of the vibration peak at resonance, as shown in Figure 8. This confirms that the GLODS framework effectively identifies optimal electrical parameters, validating both the algorithmic implementation and its coupling with the finite element model.

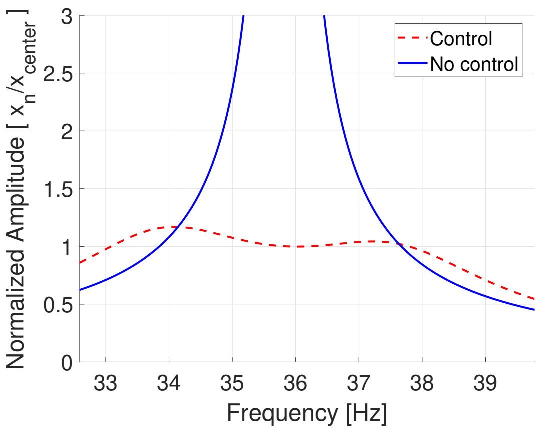


Figure 8: Initial analysis of the electric optimization model based on the maximum amplitude.

4. Results

This section presents the main results obtained from the modal and harmonic analyses of the developed numerical model. A unimodal approach was first adopted, where the first 10 natural modes of the structure were analyzed individually. For each mode, up to five pairs of piezoelectric patches were optimized using the EMCC as the performance metric. The resulting geometric configurations were then evaluated through harmonic analysis with passive single shunt circuits, whose RL parameters were calibrated according to Equation 2, to assess the vibration attenuation performance across low and high-frequency modes.

Subsequently, a multimodal optimization approach was implemented to target higher-frequency vibrations. In this case, multiple modes (6, 7 and 8) were considered simultaneously to determine optimal patch distributions, followed by fine-tuning of the electrical parameters through optimization techniques. Finally, the results from both unimodal and multimodal strategies were compared, highlighting their relative efficiency, advantages, and practical applicability for passive vibration control in smart structures.

4.1. Unimodal Approach

This section presents the results of the unimodal approach, which focuses on the individual analysis of each vibration mode of the structure. The method aims to identify the optimal configurations of piezoelectric patch pairs that maximize the electromechanical coupling coefficient ($k^2_{e(m)}$), thereby enhancing the efficiency of passive vibration control through RL shunt circuits.

The optimization process was carried out for the first ten natural modes, evaluating configurations with one to five pairs of piezoelectric patches. The results, illustrated in Figure 9, showed that configurations with one to three patch pairs achieved the highest $k^2_{e(m)}$ values, while configurations with four or more pairs exhibited reduced coupling efficiency. This reduction is attributed to the decrease in individual patch area, which diminishes their influence on the structural response.

The spatial distributions of the optimal patch configurations for representative low- and mid-frequency modes (modes 1 and 5) further illustrate how patch placement aligns with the modal displacement profiles. In the case of mode 5, however, the optimization results revealed lower coupling values, mainly due to the sensitivity of this mode to mass redistribution effects caused by the added patches. This observation highlights the influence of modal characteristics on the achievable coupling efficiency.

From the analysis of the optimal configurations, the RL parameters of the passive shunt circuits were

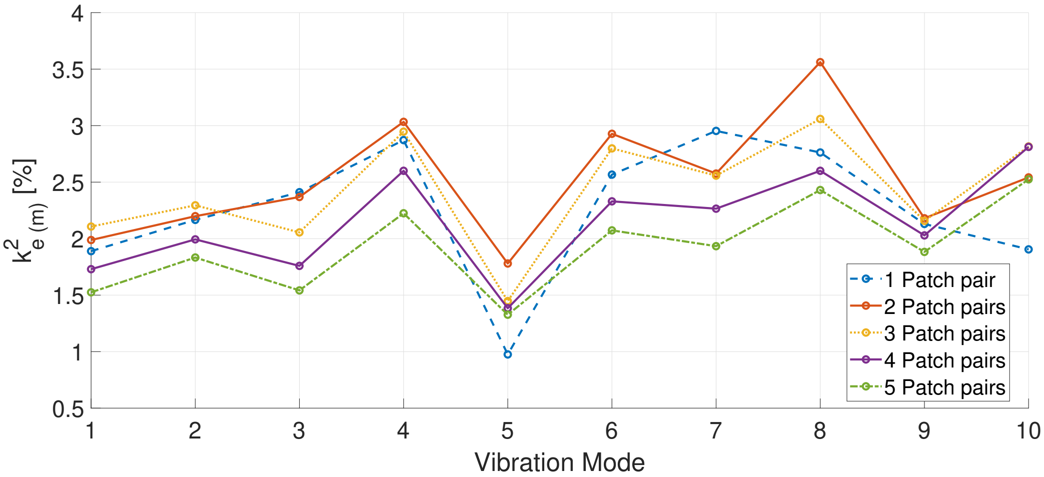


Figure 9: Distribution of the values of $k^2_{e(m)}$ related to the first 10 natural modes, with pairs of patches positioned in their respective optimal configurations for each analyzed quantity.

calibrated based on Equation 2. These data indicate that the optimized configurations for one to three patch pairs produced consistent RL parameter values across different modes, confirming that well-positioned patches yield similar electromechanical performance.

In the harmonic analysis stage, the optimal unimodal configurations were evaluated under harmonic excitation around each mode's natural frequency. The results demonstrated effective vibration attenuation for low-frequency modes (1st-3rd), where the configuration with a single pair of patches achieved the highest reduction in peak amplitude. This outcome suggests that, for low-frequency vibrations with well-separated modes, simple unimodal control remains highly efficient.

For higher-frequency modes, however, the unimodal shunt approach proved insufficient. The response of the 7th mode revealed significant interference from neighboring modes due to their close frequency spacing, as well as differences in modal charge distributions, as shown in Figure 10 using 2 patch pairs. Consequently, while the unimodal optimization performed well for isolated modes, its applicability becomes limited in multimodal regions. These findings justify the need for a multimodal optimization strategy and more advanced control configurations, which are explored in the following section.

4.2. Multimodal Approach

After validating the unimodal cases, the optimization framework was extended to a multimodal approach, aiming to simultaneously attenuate vibrations from multiple structural modes. In this stage, the DMS algorithm was employed to determine the optimal placement of the piezoelectric patch pairs

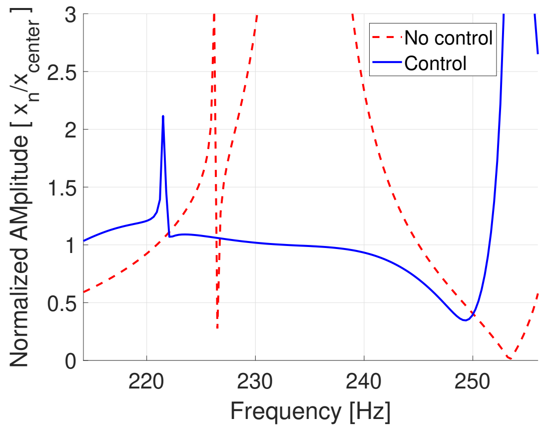


Figure 10: Harmonic solutions for 2 patch pairs and the 7$^{\text{th}}$ vibration mode using a single shunt circuit with calibrated RL values.

by maximizing the combined EMCC across selected modes. Subsequently, the GLODS algorithm optimized the electrical parameters of the RL shunt circuits, ensuring that each patch pair was tuned to different frequency regions to achieve broader control performance.

This integrated optimization enables the system to distribute the control effort efficiently across the structure, enhancing its capability to mitigate vibration peaks over a wider frequency spectrum. The resulting configurations were evaluated through harmonic analysis to assess their vibration attenuation effectiveness under different multishunt setups.

As shown in Figure 11, increasing the number of piezoelectric patch pairs significantly enhances vibration attenuation across the frequency range an-

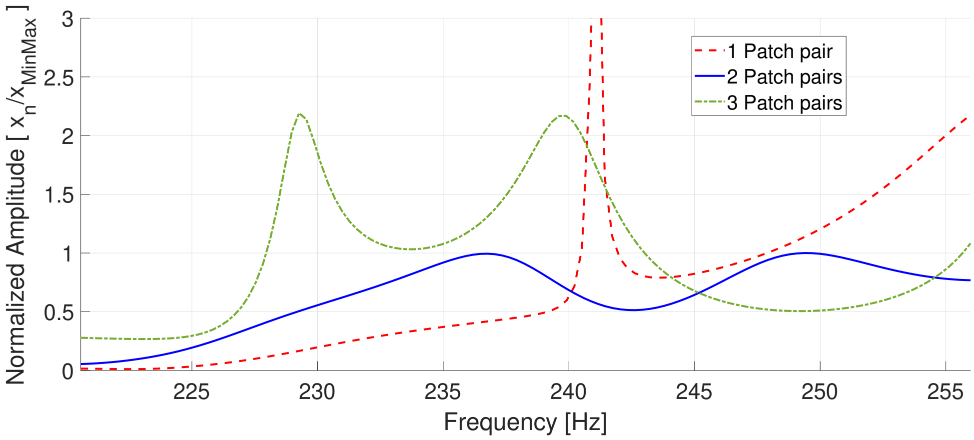


Figure 11: Comparison of the system harmonic responses considering the following electrical and geometric configurations: optimized single-shunt with 1 patch pair, optimized multi-shunt with 2 patch pairs, and optimized multi-shunt with 3 patch pairs.

alyzed. This improvement is primarily due to the use of multi-shunt configurations, which allow each patch pair to be individually tuned to a specific vibration mode, providing targeted damping for multiple resonances.

Interestingly, while the configuration with two patch pairs achieved superior vibration reduction compared to the single-pair case, the performance did not continue to improve with three pairs. This unexpected result suggests that the additional mass and altered stiffness distribution introduced by the extra patches may have adversely affected the system's overall dynamic behavior, reducing control efficiency.

Overall, the multimodal optimization results confirm the robustness of the proposed framework, demonstrating that the coordinated optimization of geometric placement and electrical tuning effectively extends the vibration control bandwidth while maintaining computational efficiency.

4.3. Comparison Between the Approaches

The comparison between the unimodal and multimodal control strategies highlights the distinct behaviors and trade-offs of each method when applied to smart structures with different numbers of piezoelectric patch pairs. In the unimodal approach, the optimization focuses on maximizing the EMCC of a single dominant mode, leading to precise attenuation of a specific vibration peak through single-shunt control. Conversely, the multimodal approach employs the combined optimization of patch placement (DMS) and electrical tuning (GLODS) to simultaneously mitigate multiple vibration modes using independent multi-shunt circuits.

As shown in Figure 12, the configuration with two piezoelectric patch pairs and multimodal optimization achieved the most balanced and effective vibration attenuation, combining the lowest peak amplitude with stable response across the entire frequency range. This superior performance results from the ability of the multi-shunt configuration to distribute control actions among different regions of the frequency spectrum, ensuring broader damping coverage.

In contrast, the unimodal single-shunt configurations with one and three patch pairs exhibited more localized control effects. The three-pair case, although providing some attenuation, did not surpass the two-pair multimodal configuration, likely due to the added mass and stiffness imbalance introduced by additional patches. These structural effects altered the natural frequencies and reduced the overall control efficiency, emphasizing the sensitivity of piezoelectric systems to mass distribution.

Overall, the results confirm that multimodal optimization with two patch pairs represents the optimal balance between control performance, computational efficiency, and structural integrity. This approach effectively extends the vibration suppression bandwidth while avoiding excessive complexity or mass loading, demonstrating its potential as a robust design strategy for passive vibration control in smart structures.

5. Conclusions

This thesis investigated passive vibration control in composite structures using piezoelectric patches connected to single- and multi-shunt circuits. Single-shunt control, based on a unimodal approach, effectively attenuates vibrations when only one dominant mode is excited, with one patch pair generally sufficient. Increasing the number of patch pairs beyond three reduces control efficiency due to decreased effective modal charge and altered en-

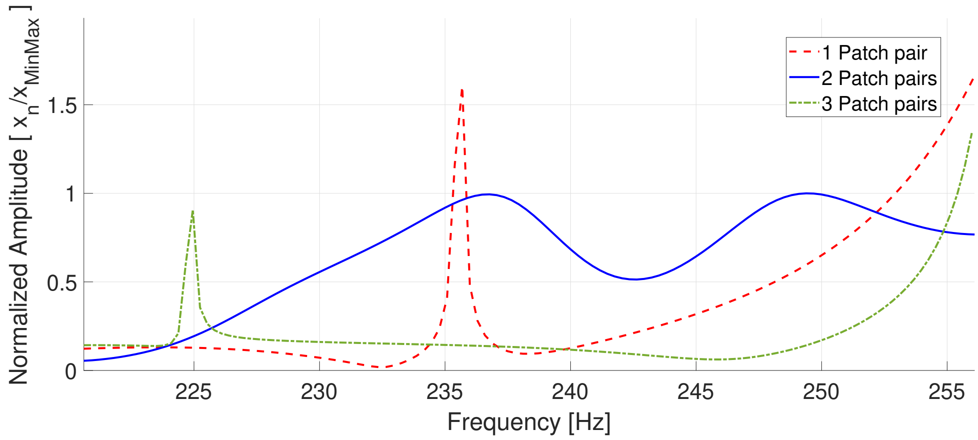


Figure 12: Comparison of the best controls implemented for each configuration of patch pair numbers: single-shunt control (unimodal approach) with 1 patch pair, multi-shunt control (multimodal approach) with 2 patch pairs, and single-shunt control (unimodal approach) with 3 patch pairs.

ergy transfer, highlighting the limitations of this approach for multi-modal scenarios.

Multi-shunt control, implemented using a multimodal approach, demonstrated superior performance when multiple vibration modes are present. Independent circuits allow selective tuning for specific regions of the frequency spectrum, improving both amplitude reduction and vibrational energy dissipation. The study also revealed that patch distribution affects the system's dynamic response: in configurations with three or more patch pairs, mass redistribution can influence peak amplitudes and may mask the actual effectiveness of the control. An energy-based assessment confirmed that, despite slightly higher peak amplitudes in some cases, the multimodal multi-shunt approach provides more robust and globally efficient vibration attenuation.

For future work, integrated optimization strategies are suggested, simultaneously considering patch positions and electrical parameters. Although computationally demanding, such methods could overcome the limitations observed with sequential modal and harmonic analyses, further enhancing the performance and robustness of passive multi-shunt control in complex structural systems.

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